Luiss
Libera Università Internazionale
degli Studi Sociali Guido Carli

Lab on Smart Contracts

Data Privacy and Security

Data Science and Management (DASMA)

LUISS

What is a Smart Contract?

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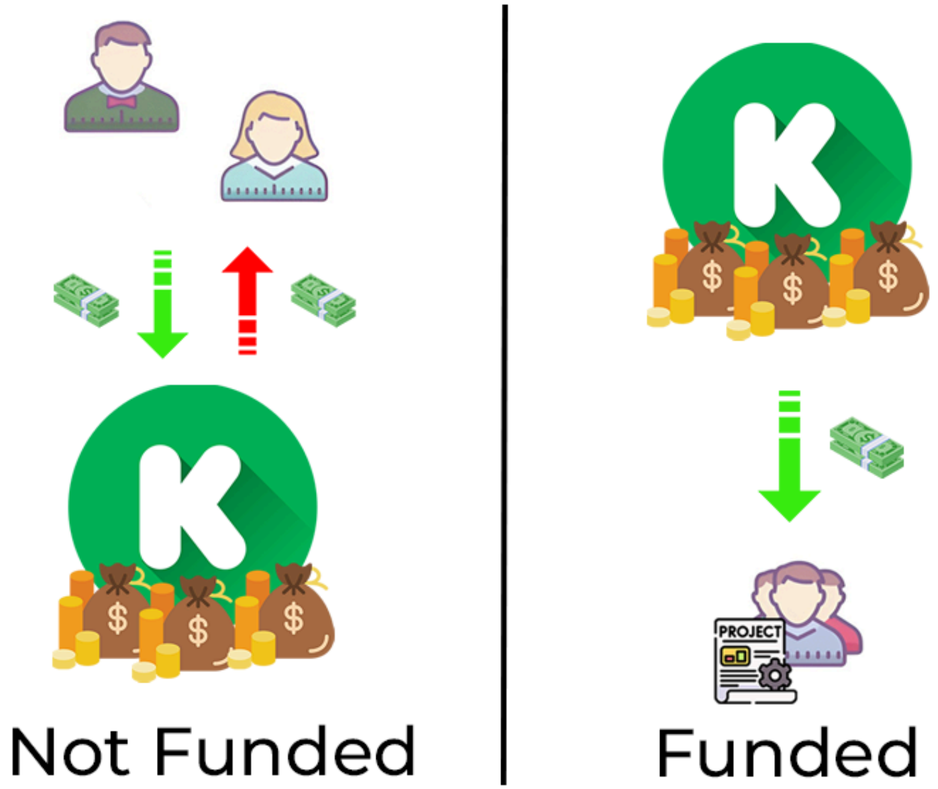
Example: consider Kickstarter - large fundraising platform for independent projects.

Both parties must trust it to handle their money correctly:

- Project successfully funded - the project team expects Kickstarter to release them the funds
- Funding goal not reached - supporters expect Kickstarter to refund them

Risks of relying on a third party:

- Mismanagement
- Technical issues
- Hacking
- ...



What is a Smart Contract?

> It is a **computer program** stored on the blockchain that executes automatically when conditions are met.

With smart contracts, we can replicate Kickstarter's model without any third party involvement:

- Program a smart contract to hold all received funds until a funding goal is reached
- Supporters send their funds directly to the smart contract:
  - If the goal is met, the contract automatically transfers the funds to the project creator
  - If the goal is not met, the funds are automatically returned to each supporter

Since smart contracts live on a blockchain, the system is fully decentralized → no single entity controls the funds.

```
contract Crowdfund {

    address creator;
    uint goal;
    uint totalFunds;
    mapping(address => uint) contributions;

    function contribute() {
        contributions[msg.sender] += msg.value;
        totalFunds += msg.value;
    }

    function finalize() {
        if (totalFunds >= goal) {
            // goal met → pay creator
            send(creator, totalFunds);
        } else {
            // goal not met → refund
            send(msg.sender,
contributions[msg.sender]);
        }
    }
}
```

Why can we trust Smart Contracts?

Because smart contracts are stored on a blockchain, they inherit some of its key properties:

- **Immutable**: once deployed, the code cannot be changed
- **Deterministic**: same inputs always produce the same outputs
- **Decentralized**: runs on all nodes of the network simultaneously
- **Atomic**: transactions either fully succeed or fully revert (no partial execution)
- **Transparent**: the code and state are publicly visible

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> That said, smart contracts are only as trustworthy as the code they're written with.

Solidity: The Language of Ethereum

Solidity is the de facto standard programming language for Ethereum smart contracts.

- Imperative, object-oriented, inspired by JavaScript/C++
- Compiled into **EVM bytecode** and deployed on the blockchain
- **Turing-complete** (unlike Bitcoin's Script language)
- In continuous evolution (frequent new versions)

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```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract SimpleStorage {
    uint256 public storedValue;

    function set(uint256 _value) public {
        storedValue = _value;
    }

    function get() public view returns (uint256) {
        return storedValue;
    }
}
```

DApps: Decentralized Applications

A **DApp** (Decentralized Application) combines:

- A **smart contract** backend (deployed on a blockchain e.g. Ethereum)
- A **web frontend** (JavaScript/HTML, running in the browser)
- A **wallet** (e.g., MetaMask) to sign transactions

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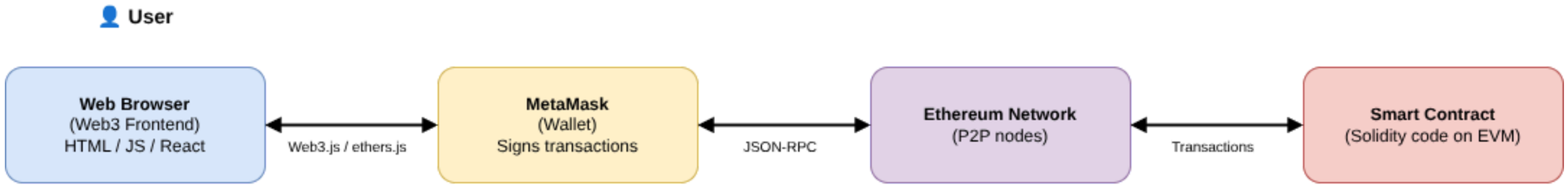

The user interacts with the frontend, but all state changes happen on the smart contract via signed transactions

How it works: the frontend talks to the smart contract through a library (e.g., ethers.js), and the wallet handles the user's identity and transaction signing.

Smart Contract Example: A Simple Faucet

A faucet is a contract that dispenses small amounts of ETH to anyone who asks:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract Faucet {
    // Accept any incoming amount
    receive() external payable {}

    // Give out ether to anyone who asks
    function withdraw(uint withdrawAmount) public {
        // Limit withdrawal amount
        require(withdrawAmount <= 0.1 ether);

        // Send the amount to the requester
        payable(msg.sender).transfer(withdrawAmount);
    }
}
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Key elements:

- `receive()`: allows the contract to accept ETH
- `msg.sender`: the address of whoever called the function
- `require()`: reverts the transaction if the condition is false
- `payable(...).transfer(...)`: sends ETH to an address

Smart Contract Lifecycle

1. **Development:** write the contract in Solidity (e.g., using the Remix web IDE at remix.ethereum.org)
2. **Compilation:** the Solidity code is compiled into EVM bytecode
3. **Deployment:** the bytecode is sent as a transaction to the zero address (`0x0`), which triggers the network to create the contract and assign it a **unique address**.
4. **Interaction:** users and other contracts can invoke its functions by sending transactions (state-changing) or calls (read-only).

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Once deployed, the code is **immutable** – bugs cannot be fixed. This makes **security auditing** critical before deployment.

Smart Contract Security Concerns

Smart contracts handle real money, so bugs can be **catastrophic**.

Common vulnerabilities:

- **Re-entrancy**: a malicious contract calls back into the victim contract before the first call finishes (e.g., the DAO hack, 2016 – $60M stolen)
- **Integer overflow/underflow**: arithmetic errors in older Solidity versions
- **Unchecked external calls**: assuming a call succeeded when it didn't
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> What makes this dangerous is immutability – once a vulnerable contract is deployed, it stays vulnerable forever.

Interacting with Smart Contracts

Block Explorers: Etherscan

A **block explorer** is a web interface to read blockchain data without running a full node.

Etherscan (https://etherscan.io) is the most popular Ethereum block explorer:

- Search for **addresses**, **transactions**, **blocks**, or **contracts**
- View transaction history, balances, and token holdings
- Read and write to **verified smart contracts** directly from the browser
- View contract source code (if verified by the developer)

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For testnets (Sepolia, Goerli), use:

- https://sepolia.etherscan.io
- https://goerli.etherscan.io

Calling Contracts Programmatically

You can interact with contracts using libraries like **web3.py** (Python) or **ethers.js** (JavaScript).

Python example:

```python
from web3 import Web3
import base64, json

w3 = Web3(Web3.HTTPProvider("https://eth-sepolia.g.alchemy.com/v2/YOUR_KEY"))

# Minimal ABI - only the function you need
abi = [{"inputs":[{"name":"tokenId","type":"uint256"}],
        "name":"tokenURI","outputs":[{"name":"","type":"string"}],
        "stateMutability":"view","type":"function"}]

contract = w3.eth.contract(address="0x...", abi=abi)
uri = contract.functions.tokenURI(0).call()
print(uri)
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The ABI (Application Binary Interface) describes the contract's functions and is needed to encode/decode calls.

NFT Metadata: `tokenURI`

ERC-721 NFTs store metadata (name, description, image, attributes) via the `tokenURI(uint256 tokenId)` function.

The `tokenURI` function returns a **URI** (Uniform Resource Identifier) pointing to the metadata:

- **IPFS**: `ipfs://Qm...` – decentralized storage
- **HTTP**: `https://example.com/metadata/123.json` – centralized server
- **Data URI**: `data:application/json; base64,...` – metadata encoded directly on-chain

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The metadata is typically JSON:

```
{
  "name": "My NFT #123",
  "description": "...",
  "image": "ipfs://...",
  "attributes": [
    {"trait_type": "Color", "value": "Blue"},
    {"trait_type": "Rarity", "value": "Legendary"}
  ]
}
```

RPC Providers: Connecting to the Blockchain

To interact with Ethereum from your code, you need to connect to a **node** that has a copy of the blockchain.

A **Web3.HTTPProvider** (or similar) is a connection to an Ethereum node via HTTP/HTTPS:

```
w3 = Web3(Web3.HTTPProvider("https://eth-sepolia.g.alchemy.com/v2/YOUR_KEY"))
```

This URL points to an **RPC endpoint** (Remote Procedure Call) – a server that accepts requests like:

- `eth_call` – read contract state (no transaction)
- `eth_sendRawTransaction` – submit a signed transaction
- `eth_getBalance` – check an address's ETH balance
- `eth_getTransactionReceipt` – check if a transaction succeeded

Why is it needed?

Running your own Ethereum node requires:

- Downloading the entire blockchain (1TB for mainnet)
- Keeping it synced 24/7
- Significant CPU, RAM, and bandwidth

Most developers use **RPC providers** instead:

- **Alchemy** (https://alchemy.com) – free tier available
- **Infura** (https://infura.io) – popular, free tier
- **QuickNode**, **Ankr**, **Chainstack** – alternatives

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The provider acts as a **gateway** to the blockchain: your code sends JSON-RPC requests, the provider's node executes them and returns results. You get all the benefits of a full node without running one yourself.

The Frontend is NOT a Security Boundary

A critical principle in blockchain security: **never trust the frontend**.

Any restrictions enforced only in JavaScript (client-side) can be trivially bypassed:

- **Input validation** (e.g., "max 100 tokens") → send 1000 directly to the contract
- **Button disabling** (e.g., "you can only vote once") → call the vote function multiple times

If the smart contract doesn't enforce a rule, **the rule doesn't exist**.

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> *The blockchain doesn't care about your UI. Only the smart contract logic matters.*

Time to Cook

CryptoFantaSanremo

Blockchain Challenge 02

CryptoFantaSanremo - Reconnaissance

CryptoFantaSanremo (500 points) blockchain

You can play FantaSanremo on the blockchain!

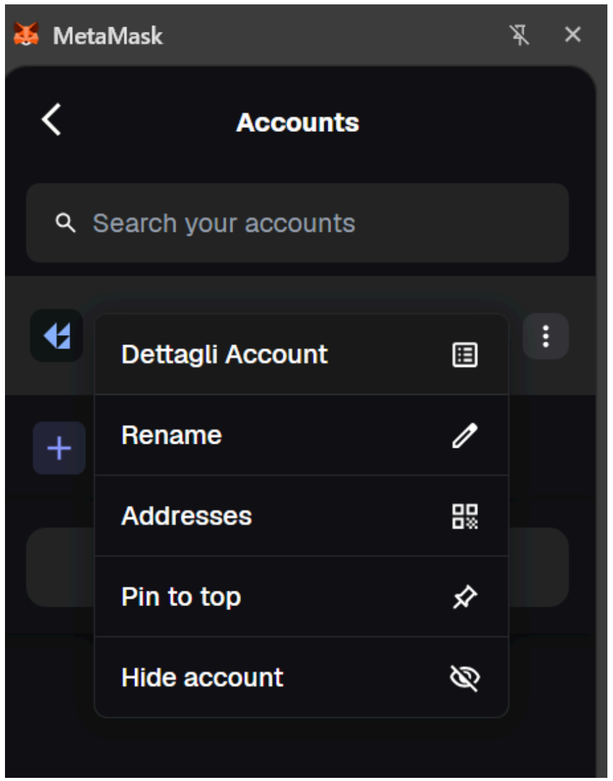
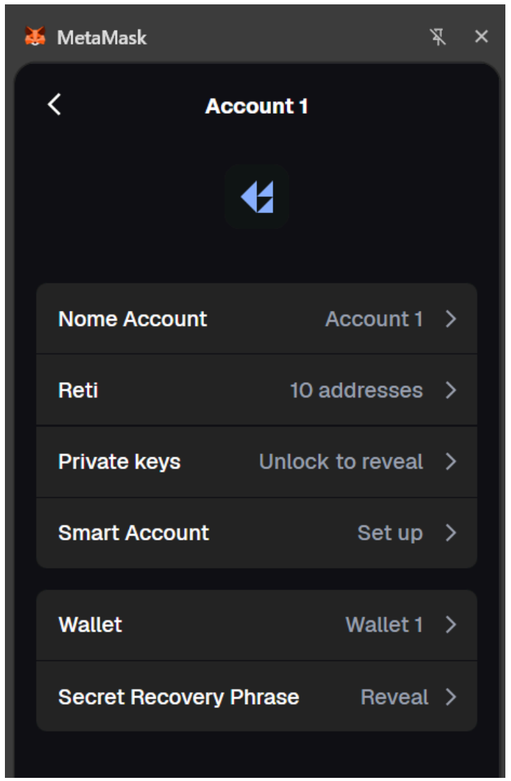
https://eth02.webhack.it

Flag

Submit

What can we observe? The challenge...

- presents a fantasy league DApp where players pick a team of 5 singers and register via a smart contract;
- exposes the full contract source code, the ABI, and ready-to-use Python scripts directly from the interface;
- awards a Flag NFT to any player whose team score reaches the threshold.

MetaMask - Private Keys

Go to Account → Account Details → Private Keys